

Geomorphology & Oceanography

Landforms, plate tectonics, earthquakes/volcanoes and ocean currents - map/concept mix.

Portion focus ~72% · Topic: geo-physical

PYQ themes: Plate tectonics · Earthquakes · Landforms · Ocean currents · Coral reefs

Why this portion (PYQ lens)

Portion chance ~72%. Recurring; parent geo-physical ~79%.

Years appeared (synthesis): 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025

Why UPSC asks this

Process - landform matching and current maps are durable PYQ styles.

Syllabus framing

Portion chance ~72%. Recurring; parent geo-physical ~79%. This portion sits inside geo-physical and centres on geomorphology and oceanography. Process - landform matching and current maps are durable PYQ styles.

Recurring PYQ angles: Plate tectonics, Earthquakes, Landforms, Ocean currents, Coral reefs. Approximate appearance years in the 2016-2025 educational synthesis: 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025. Use the chance percentage only to sequence study hours - it is not an official prediction.

Core map of geomorphology and oceanography

Plate boundaries: Convergent/divergent/transform - earthquake/volcano logic. Himalayan orogeny: Collision continuum - seismic implications.

Weathering vs erosion: In-situ decay vs transport removal. Fluvial landforms: Meanders, oxbow, deltas, terraces.

Hold this core map as your elimination spine before memorising peripheral CA overlays.

Core concepts and definitions

Glacial landforms: Cirque, moraine - limited India but asked. Coastal landforms: Bars, lagoons, cliffs, mangroves interface.

Karst: Limestone solution landscapes. Earthquake focus/epicentre: Definitions.

Institutions, processes and mechanisms

Tsunami generation: Displacement of water column. Volcanic types: Shield vs composite outline.

Ocean currents: Warm/cold; climate influence on coasts. Upwelling: Nutrient-rich fisheries link.

India angle and applied examples

Tides: Gravitational; spring/neap idea. EEZ vs territorial sea: UNCLOS vocabulary bridge.

Coral reefs: Conditions: warm, clear, shallow - bleaching CA. Continental shelf: Resource and hazard relevance.

Comparisons, linkages and discriminators

Cyclone ocean heat: Warm SST energy source. Seamounts/trenches: Map terms.

Mass movements: Landslides in Himalaya - DM bridge. Isostasy outline: Vertical adjustment idea.

India angle

Himalayan seismicity + Indian Ocean tsunami literacy + coral bleaching headlines.

Question patterns and exam spine

Matching lists and statements on geomorphology and oceanography.

Revision spine: Definitions -> table -> India -> traps.

Eliminate anachronisms and category errors first, then test remaining options against table rows and must-remember triggers.

Trap watch-list (specific)

Avoid: All plate edges identical hazards. Avoid: Cold currents always dry coasts absolutely. Avoid: Coral in cold deep seas. Avoid: Focus=epicentre.

Boundary hazards

Boundary | Typical association

Convergent -- Quakes/volcanoes/orogeny

Divergent -- Rifting volcanism

Transform -- Strike-slip quakes

Map / list facts

- Ring of Fire contrast
- Indian Ocean currents sketch
- Andaman trench awareness

Common traps

- x All plate edges identical hazards
- x Cold currents always dry coasts absolutely
- x Coral in cold deep seas
- x Focus=epicentre

Must remember

- 3 boundary types
- Weathering != erosion
- Warm vs cold currents idea
- Coral conditions
- Focus vs epicentre

Quick revision checklist

- [] Reproduce table
- [] 5 must-remembers
- [] 3 traps
- [] India anchor
- [] 10 MCQs

Educational prioritisation heuristic - not an official UPSC prediction. (c) UPSC IAS Study Hub original notes.